More Circuits: Ammeters & Voltmeters (completed)

Goal.

- Meet two measuring devices — the ammeter and the voltmeter — and figure out from first principles what resistance each one must have.
- See that the loop rule is bigger than "find the current": the same equation lets you solve for *whatever single thing* you don't already know, once a meter tells you something new.
- Work three versions of one loop, each with a different unknown.

1. Ideal meters

Ammeter: measures current. Inserted in series in the circuit (break the wire, wire the meter into the gap) so the same current that was already flowing still flows straight through it. An ideal ammeter has resistance 0.

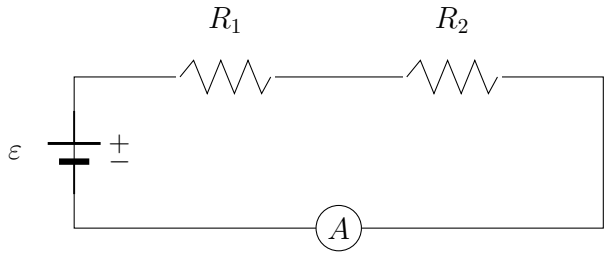
Voltmeter: measures the potential difference (ΔV) between two points. Connected in parallel across an element (its own branch, tapping both ends — it is never spliced into the main wire). An ideal voltmeter has resistance ∞ (essentially infinite).

Check. An ammeter must not change the current it's reporting; a voltmeter must not divert any current away from the branch it's tapping. Given that, which meter needs zero resistance, and which needs (essentially) infinite resistance?

The ammeter needs $R = 0$: any resistance of its own would add an extra IR drop and throttle the very current it's trying to report. The voltmeter needs $R \to \infty$: a finite resistance would give current an extra path to leak down, siphoning some of it away from the element being measured. Zero resistance in series, infinite resistance in parallel — both add nothing to the loop-rule sum.

2. Problem 1 — the current

A 9.0 V ideal battery drives current through $R_1 = 1.8$ kΩ and $R_2 = 2.7$ kΩ in series. An ammeter is wired into the loop as shown.



Write the loop rule for this circuit and solve for the current the ammeter reads.

Loop rule (the ammeter, $R = 0$, contributes no drop of its own):

$$\varepsilon - IR_1 - IR_2 = 0 \quad \Longrightarrow \quad \varepsilon = I(R_1 + R_2).$$

$$\boxed{I = \frac{\varepsilon}{R_1 + R_2}}$$

Now with the numbers above:

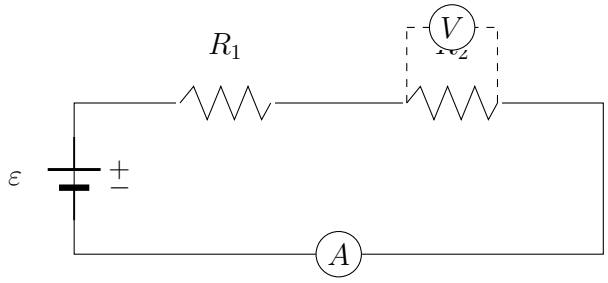
$$I = \frac{9.0\ \text{V}}{1.8\ \text{k}\Omega + 2.7\ \text{k}\Omega} = \frac{9.0\ \text{V}}{4.5\ \text{k}\Omega} = 2.0\ \text{mA}$$

Check. The ammeter sits on the bottom wire, not between R_1 and R_2. Would it read something different if you moved it anywhere else in this loop? Why?

No — this is a single series loop, so the series rule already says the current is the same everywhere around it. An ideal ammeter adds no resistance of its own, so sliding it to any other point in the loop reports that same 2.0 mA.

3. Problem 2 — a voltmeter's reading

Same loop, but now only $R_1 = 2.0\ \text{k}\Omega$ and the EMF ($\varepsilon = 12\ \text{V}$) are given — R_2's resistance is unknown. The ammeter reads $I = 3.0\ \text{mA}$, and a voltmeter is connected across R_2.



Use the loop rule to predict the voltmeter's reading. (You will not need R_2's resistance for this part.)

Loop rule — the voltmeter's branch carries no current (ideal, $R = \infty$), so it just reports whatever ΔV_2 the loop rule assigns to that gap:

$$\varepsilon - IR_1 - \Delta V_2 = 0.$$

$$\boxed{\Delta V_2 = \varepsilon - IR_1}$$

Now with the numbers above:

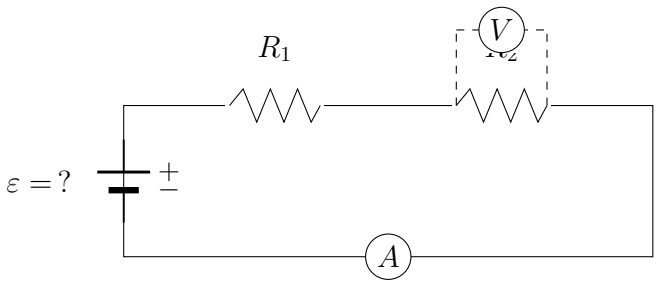
$$\Delta V_2 = 12\ \text{V} - (3.0\ \text{mA})(2.0\ \text{k}\Omega) = 12\ \text{V} - 6.0\ \text{V} = 6.0\ \text{V}$$

Check. You found the voltmeter's reading without ever using R_2's resistance. Why didn't you need it?

The loop rule only cares about voltage *drops*, not about what causes them. Once you know ε and the drop across R_1, the remaining drop around the loop is fixed by energy conservation alone — whatever resistor (or combination) sits there, it must drop exactly 6.0 V to make the loop sum to zero.

4. Problem 3 — the EMF itself

Same loop again, but now the battery's EMF is what's unknown. $R_1 = 2.0\ \text{k}\Omega$ is given, the ammeter reads $I = 4.0\ \text{mA}$, and the voltmeter across R_2 reads 5.0 V.



Use the loop rule to find ε.

Same loop rule, just solved for a different letter this time:

$$\varepsilon - IR_1 - \Delta V_2 = 0.$$

$$\boxed{\varepsilon = IR_1 + \Delta V_2}$$

Now with the numbers above:

$$\varepsilon = (4.0\ \text{mA})(2.0\ \text{k}\Omega) + 5.0\ \text{V} = 8.0\ \text{V} + 5.0\ \text{V} = 13.0\ \text{V}$$

Check. Using I and the voltmeter reading, could you now find R_2 too, even though its resistance was never given directly? What is it?

Yes — Ohm's law on R_2 alone: $R_2 = \Delta V_2 / I = 5.0\ \text{V}/4.0\ \text{mA} = 1.25\ \text{k}\Omega$. Across all three problems, the loop rule plus one meter reading was always enough to pin down whichever single quantity was missing.

5. Two loops, two batteries

A single loop only ever gives you *one* equation. The circuit below has three branches meeting at b and at c, so there are three unknown currents I_1, I_2, I_3 — one loop equation can't pin down three unknowns.

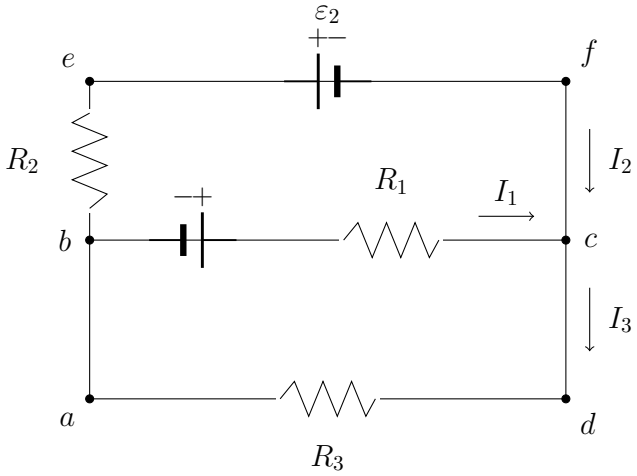
How many equations do you need, and where does each one come from? Three: pick any *two* of the loop's windows (a third loop is just their sum, not new information) and add the junction rule at b or c.

Check. Why doesn't the loop rule applied to the *outer* perimeter (e-f-d-a-e) give you a fourth, independent equation?

Walking the outer perimeter is the same as walking Loop 1 and Loop 2 back to back — the shared b-c branch gets traversed once each way, so its terms cancel and you're left with exactly Loop 1 + Loop 2 added together. It's a true statement, just not an *independent* one.

6. Problem 4 — all three currents

In the circuit below, $\varepsilon_1 = 9.0$ V, $R_1 = 4\ \Omega$, $\varepsilon_2 = 16$ V, $R_2 = 5\ \Omega$, $R_3 = 2\ \Omega$.



Write the loop rule for each loop and the junction rule at c, then solve the resulting system for all three currents.

$$I_1R_1 - I_2R_2 = \varepsilon_1 + \varepsilon_2 \quad \Rightarrow \quad 4I_1 - 5I_2 = 25$$
$$I_1R_1 + I_3R_3 = \varepsilon_1 \quad \Rightarrow \quad 4I_1 + 2I_3 = 9$$
$$I_1 + I_2 = I_3$$

Substituting $I_3 = I_1 + I_2$ into the second equation gives $4I_1 + 2(I_1 + I_2) = 9 \Rightarrow 6I_1 + 2I_2 = 9$. Combined with $4I_1 - 5I_2 = 25$, eliminate I_2: multiply the first by 2 and the second by 5, then add:

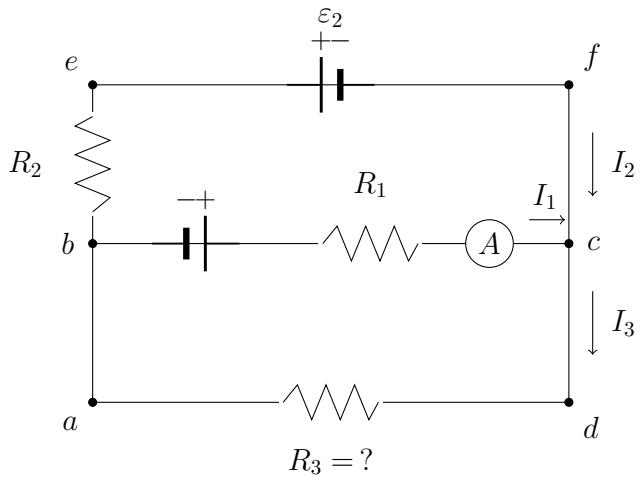
$$I_1 = 2.5\ \text{A} \qquad I_2 = -3.0\ \text{A} \qquad I_3 = I_1 + I_2 = -0.5\ \text{A}$$

Check. One of your currents came out negative. What does that tell you about the direction you assumed for it on the diagram?

Both I_2 and I_3 came out negative — the actual current in each of those branches flows opposite to the arrow drawn. I_1 came out positive, so that arrow was right. Check: $I_1R_1 - I_2R_2 = 2.5(4) - (-3.0)(5) = 10 + 15 = 25 = \varepsilon_1 + \varepsilon_2$, and $I_1R_1 + I_3R_3 = 2.5(4) + (-0.5)(2) = 10 - 1 = 9 = \varepsilon_1$. ✓

7. Problem 5 — an unknown resistor

Same shape, new numbers: $\varepsilon_1 = 20$ V, $R_1 = 3\ \Omega$, $\varepsilon_2 = 10$ V, $R_2 = 5\ \Omega$. This time R_3 is unknown — but an ammeter in the ε_1-R_1 branch reads $I_1 = 4.0$ A.



Use the loop rule and the junction rule to find R_3. (Work out I_2, then I_3, on your way there.)

Loop 1 is the only equation with ε_2 and R_2 in it, and I_1 is now known, so solve it for I_2 first:

$$I_1R_1 - I_2R_2 = \varepsilon_1 + \varepsilon_2 \ \Rightarrow\ 3(4) - 5I_2 = 20 + 10 \ \Rightarrow\ I_2 = -3.6 \text{ A}$$

Junction rule:

$$I_3 = I_1 + I_2 = 4.0 + (-3.6) = 0.4 \text{ A}$$

Loop 2 is the only equation with R_3 in it:

$$\varepsilon_1 = I_1R_1 + I_3R_3 \ \Rightarrow\ 20 = 3(4) + 0.4R_3$$

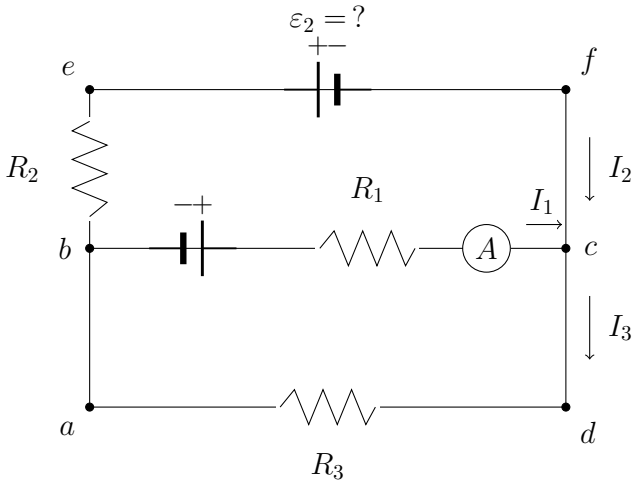
$$\boxed{R_3 = \frac{\varepsilon_1 - I_1R_1}{I_3}} \qquad R_3 = \frac{20 - 12}{0.4} = 20\ \Omega$$

Check. Could you have found R_3 without first finding I_2? Why or why not?

No. R_3 only shows up in Loop 2, and Loop 2 has two unknowns in it, I_3 and R_3, until I_3 is pinned down. I_3 only comes from the junction rule, which needs I_2 — and I_2 only comes from Loop 1 (the one equation that actually contains ε_2 and R_2). Every given number gets used along exactly one chain.

8. Problem 6 — an unknown EMF

Same shape again: $R_1 = 5\ \Omega$, $R_2 = 6\ \Omega$, $R_3 = 4\ \Omega$. This time ε_2 is unknown, but $\varepsilon_1 = 21$ V is given, and the ammeter reads $I_1 = 3.0$ A.



Find ε_2. (This time work out I_3 first, then I_2.)

Loop 2 is the only equation with R_3 in it, and it only involves I_1 and I_3 — solve it first:

$$\varepsilon_1 = I_1R_1 + I_3R_3 \ \Rightarrow\ 21 = 3(5) + 4I_3 \ \Rightarrow\ I_3 = 1.5 \text{ A}$$

Junction rule:

$$I_2 = I_3 - I_1 = 1.5 - 3.0 = -1.5 \text{ A}$$

Loop 1 is the only equation with ε_2 in it:

$$I_1R_1 - I_2R_2 = \varepsilon_1 + \varepsilon_2 \ \Rightarrow\ 3(5) - (-1.5)(6) = 21 + \varepsilon_2$$

$$\boxed{\varepsilon_2 = I_1R_1 - I_2R_2 - \varepsilon_1} \qquad \varepsilon_2 = 15 + 9 - 21 = 3.0 \text{ V}$$

Check. Compare the order you worked in for Problems 5 and 6 — which loop did you need *first* in each, and why did the unknown force that order?

Problem 5's unknown (R_3) lives only in Loop 2, but Loop 2 was stuck with two unknowns (I_3 and R_3) until I_2 — found from Loop 1 — fed the junction rule. Problem 6's unknown (ε_2) lives only in Loop 1, but Loop 1 needed I_2, which needed I_3 from Loop 2 first. Whichever loop *doesn't* contain the unknown is always the one you solve first — it's the one that's already fully determined by what you're given.